

GWALIOR, India

WMO: 423610

Lat: 26.2112N

Lon: 78.2024E

Elev: 617

StdP: 14.37

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.7	44.6	32.4	27.5	73.0	35.8	31.5	67.1	10.2	65.1	7.8	64.9	0.2	140	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.8	110.8	74.2	108.7	74.0	106.3	74.0	83.6	92.5	82.8	91.4	82.1	90.4	4.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	168.7	87.5	80.9	164.2	86.7	80.2	160.7	86.1	48.5	92.8	47.6	91.7	46.8	90.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	9.8	7.7	DB	39.1	113.9	2.6	1.7	37.2	115.2	35.7	116.2	34.2	117.2	32.3	118.4	(4)
(5)			WB	37.9	85.6	2.4	1.1	36.2	86.3	34.8	86.9	33.4	87.5	31.7	88.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.1	59.0	66.5	77.1	87.4	94.0	94.1	87.4	85.1	84.7	80.1	71.1	62.4		
	DBStd	12.26	4.78	5.08	5.55	4.95	4.17	5.58	4.21	2.78	2.85	3.93	4.00	4.63		
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	343	196	38	1	0	0	0	0	0	0	0	3	105		
	CDD50	10634	280	461	840	1123	1365	1322	1160	1088	1042	934	634	385		
	CDD65	5500	9	79	376	673	900	872	695	623	592	469	187	25		
	CDH74	75242	130	870	4337	9841	14929	14139	9468	7758	7158	4748	1594	270		
	CDH80	44263	14	259	2345	6414	10591	9858	5073	3441	3302	2387	549	30		
(14) Wind	WSAvg	1.8	0.9	1.7	1.8	2.2	2.9	3.1	2.9	2.5	2.0	1.0	0.5	0.6		
(15) Precipitation	PrecAvg	31.3	0.4	0.6	0.4	0.2	0.5	3.1	9.3	9.0	6.4	0.8	0.2	0.5		
	PrecMax	43.1	2.3	4.2	4.8	1.1	1.8	9.8	15.2	16.1	21.2	5.1	3.2	5.5		
	PrecMin	20.0	0.0	0.0	0.0	0.0	0.0	0.1	0.9	2.2	0.9	0.0	0.0	0.0		
	PrecStd	7.6	0.6	0.9	0.9	0.3	0.4	2.6	3.4	3.7	4.4	1.3	0.7	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.8	91.2	102.3	109.7	113.0	114.0	105.7	98.4	98.4	98.0	91.7	83.5		
		MCWB	63.5	66.2	70.6	71.0	74.5	73.3	78.2	79.0	76.6	72.6	67.6	64.3		
	2%	DB	77.6	86.8	98.5	107.2	111.1	111.1	101.6	94.9	96.2	95.7	88.4	80.1		
		MCWB	61.3	65.9	67.6	70.2	73.7	75.6	79.3	80.4	77.9	72.9	67.1	62.3		
	5%	DB	74.0	82.9	95.3	104.8	109.3	108.5	97.9	92.8	94.4	93.7	86.0	77.1		
		MCWB	59.1	63.7	67.8	69.9	73.6	75.8	79.9	80.9	78.5	72.7	66.7	61.3		
	10%	DB	70.5	79.6	92.1	102.4	107.2	105.6	94.9	90.9	92.6	91.4	83.4	73.8		
		MCWB	57.5	62.2	67.3	70.4	73.6	76.7	80.2	80.7	78.4	72.7	65.8	60.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	70.5	75.9	79.0	83.0	84.2	84.5	84.3	84.1	81.0	73.0	67.4		
		MCDB	77.6	83.9	95.1	100.0	104.7	97.6	92.4	90.9	91.8	90.4	83.4	77.5		
	2%	WB	63.5	67.9	72.3	75.5	80.5	82.7	83.6	83.1	82.7	79.2	71.2	65.0		
		MCDB	73.9	81.7	91.4	98.9	101.6	96.3	91.4	89.8	90.3	87.8	82.4	76.2		
	5%	WB	61.0	65.9	70.2	73.5	78.4	81.7	82.8	82.4	81.6	77.4	69.3	63.2		
		MCDB	70.1	79.3	88.7	97.5	99.7	94.8	90.4	89.1	89.0	86.0	80.4	74.1		
	10%	WB	59.1	64.0	68.4	71.8	76.9	80.7	82.0	81.7	80.7	75.9	67.4	61.3		
		MCDB	67.1	76.2	86.5	95.1	98.9	93.3	89.4	87.8	87.7	85.0	79.8	71.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.8	25.2	27.1	27.7	24.8	18.4	11.8	10.6	14.3	23.8	27.2	25.2		
		MCDBR	28.2	28.9	30.8	29.6	26.7	22.2	16.1	13.6	18.2	25.7	28.7	28.5		
	5% WB	MCWBR	13.3	11.7	10.1	9.6	8.2	6.3	4.6	5.0	6.1	9.2	11.7	12.9		
		MCWBR	23.8	25.1	27.7	27.9	24.2	17.5	12.4	11.6	13.3	19.1	24.8	25.9		
(40) Clear-Sky Solar Irradiance	taub		0.498	0.475	0.490	0.560	0.636	0.701	0.693	0.619	0.563	0.599	0.591	0.555		
		taud	1.860	1.971	1.966	1.830	1.715	1.637	1.670	1.844	1.942	1.761	1.707	1.748		
	Ebn at Noon		231	252	257	242	223	208	209	225	234	213	201	206		
		Edh at Noon	57	55	58	68	76	82	79	66	58	66	66	61		
(44) All-Sky Solar Radiation	RadAvg		1119	1507	1911	2164	2195	1879	1465	1469	1610	1559	1252	1096		
	RadStd		122	98	72	68	83	142	137	121	145	89	91	62		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +0.72	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page